

Hospital Form Digitization Demo Concept

1. Scan / Upload Form

Hospital staff opens the web application and either:

- Clicks a photo of the patient's physical form
- Uploads an existing image of the form

The complete original image of the form should also be stored permanently with the patient's digital record.

2. AI Reads the Form

The AI analyzes the uploaded form and detects handwritten information from predefined fields such as:

- Patient Name
- Phone Number
- Mobile Number
- Email ID
- Age
- Date
- Blood Group
- Disease / Complaint
- Doctor Assigned
- Any other hospital-defined information

Since the hospital generally uses a fixed form format, the system can also understand where each field is normally located on the page. This should improve extraction accuracy significantly.

3. Detect Photograph, Signature & Thumb Impression

From the same uploaded form image, the AI should detect visual elements separately.

If a patient photograph is pasted on the form, the system should automatically crop it and place it inside the **Patient Photograph** section.

The same process should be followed for:

- Patient Photograph
- Patient Signature
- Thumb Impression

If any of these elements are not available on the form, the application should simply show:

Not Detected

The system should never automatically generate or assume missing information.

4. Verification Screen

After AI processing, hospital staff should see a verification screen.

Left Side

Original Uploaded Form

Right Side

Digitized Patient Information

All AI-detected fields should remain editable.

Example:

Patient Name: Rahul Sharma
Mobile Number: 98XXXXXX21
Blood Group: B+
Disease: Diabetes

Staff can compare the original handwritten form with the extracted information before saving the record.

For fields where AI confidence is low, the application can highlight them for manual review.

Example:

Rahul Sharma — 96% Confidence

Diabetes — 71% Confidence — Review Required

5. Save Patient Record

Once the information has been verified, the hospital staff clicks:

Save Patient

The application then creates a structured patient record containing:

- Patient information
- Patient photograph

- Signature
- Thumb impression
- Original scanned form
- Scan date and time
- Assigned doctor
- Patient ID / Registration Number

Live Preview

After saving the patient record, the application should display two separate preview tabs.

Doctor View

The Doctor View should work like a compact digital patient sheet.

It can contain:

- Patient Name
- Patient Photograph
- Patient ID
- Contact Information
- Age
- Blood Group
- Disease / Complaint
- Assigned Doctor
- Registration Date
- Current Appointment
- Next Appointment
- Signature / Thumb Impression
- Original Scanned Form

Later, this section can be expanded with:

- Visit History
- Prescriptions
- Medical Reports
- Doctor Notes
- Previous Appointments

Patient View

The Patient View should be much simpler and should work like a digital information receipt.

Example:

Hospital Name

Patient: Rahul Sharma

Patient ID: HSP-10238

Doctor: Dr. Amit Sharma

Visit Date: 24 August 2026

Next Appointment: 31 August 2026, 11:30 AM

General Instructions

- Take medicines as prescribed by the doctor.
- Bring previous medical reports during the next visit.
- Arrive at least 15 minutes before the scheduled appointment.
- Contact the hospital in case the appointment needs to be rescheduled.

At the bottom, provide:

Download Patient Receipt

Print Patient Receipt

A QR code can also be added later so the patient can reopen the digital receipt from their phone.

Recommended Product Workflow

The main product story should be:

Scan Form → AI Digitizes → Staff Verifies → Save → Generate Doctor & Patient View

The system should not directly push AI-extracted information into the hospital database without human verification.

This is especially important for critical information such as:

- Patient Name
- Blood Group
- Disease
- Doctor Assignment
- Phone Number

Demo Scope

The first demo should not attempt to become a complete hospital-management system.

The demo only needs to prove one major problem:

Convert a handwritten physical patient form into a structured digital patient record within seconds.

The demo can initially contain five main screens:

1. Dashboard
2. Scan / Upload Form
3. AI Processing
4. Verify Extracted Information
5. Doctor Preview / Patient Preview

Future Scalability

The form fields should be configurable instead of permanently hard-coded.

Different hospitals may have fields such as:

- Aadhaar Number
- Address
- Gender
- Emergency Contact
- Guardian Name
- Allergies
- Medical History
- Insurance Information
- Referral Doctor

The system should therefore allow different hospitals to configure their own form structure without rebuilding the complete application.

Core Value Proposition

Turn handwritten hospital registration forms into verified digital patient records within seconds while keeping the original physical form digitally archived.

Alternative Approach – Universal Form Digitization Platform

Instead of building the technology around one fixed hospital form, we can make the system completely independent from any specific form format or industry.

The same platform can then be used by:

- Hospitals
- Schools
- Colleges
- Government departments
- Clinics
- Offices
- Insurance companies
- HR departments
- Any organization that still collects information through physical forms

Sub-Admin System

Every organization will receive its own **Sub-Admin Panel**.

The organization admin can create their own digital form according to the physical form they already use.

They do not need any coding or technical knowledge.

Create a Form

The admin clicks:

Create New Form

They can give the form a name, such as:

New Patient Registration Form

Student Admission Form

Employee Joining Form

Insurance Application Form

Create Sections

The admin can divide the form into different sections.

For example:

Personal Information

Contact Information

Medical Information

Documents

Declaration

Create Fields

Inside every section, the admin can add fields from the available options.

For example:

- Short Text
- Long Text
- Name
- Phone Number
- Email
- Number
- Date
- Age
- Address
- Dropdown
- Checkbox
- Radio Button
- Photograph
- Signature
- Thumb Impression
- Document/Image
- Custom Field

The admin simply needs to look at their existing physical form and recreate every writable field inside the digital form builder.

For example, if the physical form contains:

Patient Name: _____

The admin creates:

Field Name: Patient Name

Field Type: Text

If the physical form contains:

Blood Group: _____

The admin creates:

Field Name: Blood Group

Field Type: Dropdown / Text

The same process continues until the digital structure matches the physical form.

Publish the Form

Once all the required fields have been created, the admin clicks:

Publish Form

The platform generates a unique link for that form.

Example:

Form Name: Patient Registration

Status: Published

Form Link: Generated automatically

The organization can then use this link internally on computers, tablets or mobile devices.

Scan Physical Form

Staff opens the published form link and clicks:

Scan Physical Form

They either:

- Take a picture of the physical form
- Upload an existing scanned image

The AI reads the uploaded physical form and compares the detected information with the fields created by the admin.

For example, if the digital form contains:

Patient Name

Mobile Number

Age

Blood Group

Disease

Photograph

Signature

The AI will specifically search the uploaded physical form for information corresponding to these fields.

AI Populates the Digital Form

The extracted information is automatically placed inside the respective digital fields.

Example:

Patient Name: Rahul Sharma

Mobile Number: 98XXXXXX21

Age: 42

Blood Group: B+

Disease: Diabetes

If the form contains a photograph, signature or thumb impression, AI will detect and crop them separately and place them inside the corresponding image fields.

Manual Verification

Before saving, staff receives a verification screen showing:

Original Physical Form

alongside

AI Generated Digital Form

Any incorrect information can be edited manually before final submission.

Save Record

After verification, the staff clicks:

Save Record

The system stores:

- Structured digital information
- Original scanned form
- Extracted photograph
- Signature
- Thumb impression
- Uploaded documents

- Creation date and time
- Form used
- User who processed the record

Why This Approach Is More Powerful

The technology is no longer limited to a specific hospital or a specific form.

The organization itself defines what information the AI should extract.

The platform becomes a reusable **Physical Form to Digital Data Infrastructure**.

One hospital can create a patient-registration form.

A school can create an admission form.

A company can create an employee onboarding form.

A clinic can create a medical-history form.

All of them use the same underlying technology.

Core Workflow

Admin Creates Digital Form → Publish → Staff Opens Form Link → Scan Physical Form → AI Extracts Data → Staff Verifies → Save Digital Record

Core Value Proposition

Create any digital form once and convert matching handwritten physical forms into structured digital records using AI.